

Predicting Small-Target Radar Reliability under Wave-Driven Vessel Motion

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Abstract

The ultimate goal of this work is to predict whether the next radar scan will detect a small maritime target (within an acceptable position error) and identify the physical mechanism of target occlusion when detection fails.

1 Introduction & Objective

In oceans, lakes, and other bodies of water (henceforth referred to as maritime environments), small targets can exhibit intermittent detection in radar returns. While the causes for this fading effect are numerous, the largest contributors, in no particular order, are wave geometry, radar beam motion induced by vessel pitch, and distorted scan positioning.

Clearly, it is desirable to eliminate these effects and, barring that, predict when they will occur. Here, we take advantage of the fact that the same forcing function—wavelike motion of the water surface—controls all three mechanisms. For simplicity, we assume that wind effects are not modeled independently and are represented through the wave field.

A wealth of literature already prescribes the individual mechanisms to build our model and the associated predictions. Waves and their interactions with vessels and objects are well-studied. In a similar fashion, inter- and intra-wave interactions (such as diffraction and propagation) are well-understood. The study of heave, pitch, and roll of vessels and their effects on radar scans is also well documented. Finally, the effects of radar beam distortion and general distortion against waves are also documented.

Therefore, the objective of our work is to combine these mechanisms into a single interpretable model for estimating the probability of small target detection in the subsequent radar scan. The model will primarily be tested in a synthetic environment, but an evaluation using real-world data will be included to validate the model’s predictions, relying on publicly available datasets.

2 Background & Related Works

Small maritime targets are nearly impossible to detect when their returns are comparable or weaker than surrounding sea clutter. As a solution, signal processing techniques, such as signal averaging, time-frequency analysis, and morphological filtering on real marine radar data have been utilized.

The work done by Panagopoulos and Soraghan evaluates these techniques and show that detection should be treated as an intermittent event since a target can become hidden or shadowed by waves on an interval. [1] Their work, however, makes no attempt to predict these interruptions to detections in maritime environments.

This indicates, somewhat obviously, that wave geometry changes the radar path to a target and the illumination of the sea surface. Plant and Farquharson develop some work on distinguishing geometric shadowing from partial shadowing, in which diffraction and weaker scattering can affect received signals. [2] Their measurements show that a simple geometric blockage model does NOT explain all low-grazing-angle radar behavior.

Janssen et al. further show that diffraction changes the illumination field near and beyond wave-shadow boundaries, where geometric optics can misrepresent the illumination field. Of course, their work only considers the illumination field, and not the resulting radar return, and the latter half of the detection stage must be implemented in our work. These findings motivate a geometric visibility baseline followed by a diffraction sensitivity model. [3]

Because a rotating radar acquires azimuths (or spokes; vernacular can vary between literature) at different times, its scan is formed from measurements taken at different sensor positions and orientations. Work on shipborne radar georeferencing has already studied the importance of time-tagged measurements, position, heading, and installation offsets when mapping radar imagery to geographic coordinates. [4]

Additional research on spinning radar shows that platform motion during acquisition can heavily distort a scan, requiring correction methods to be developed/utilized. Together, these studies motivate modeling the radar pose at each acquisition time and evaluating per-azimuth correction separately from losses that prevent a return from being detected. [5] The full related-work discussion appears in Appendix E.

3 Method, Software, & Data

The primary experiment will use a MATLAB simulator to generate a shared wave field and propagate its effects through radar-vessel motion, target motion, illumination, and scan geometry. If MATLAB proves to be computationally intensive, alternative implementations in C++ may be considered. The field consists of prescribed linear wave components with defined amplitudes, frequencies, directions, and phases, each according to literature data.

Each component will be evaluated at the radar vessel and target locations, preserving their spatial phase relationship. Vessel heading and speed will determine the encounter frequency. Damped roll, pitch, and heave modes will represent vessel response, while a separate heave mode will represent target motion. The governing equations and response parameters are given in Appendix A.

The received echo will combine target scattering, antenna gain, two-way propagation, sea clutter, and receiver noise. Radar cross section (RCS) will represent the target scattering strength. A fixed detector, calibrated using development data, will determine whether the echo produces a detection. This separates losses that prevent a return from being acquired from errors introduced when an acquired return is mapped to position. The echo model and ray-time geometry are detailed in Appendix B.

The primary synthetic study will use known target geometry and complete wave, vessel, and target states, allowing each failure mechanism to be isolated and compared under matched conditions. Pose correction will be evaluated separately using midpoint pose, per-azimuth horizontal-motion correction, and per-azimuth attitude correction. Supporting model calculations and measured radar evidence appear in Appendix C; the evaluation dataset remains to be selected.

4 Evaluation & Metrics

Four paired truth conditions will represent a reference case, vessel motion only, wave-path modulation only, and coupled motion and propagation. Each condition will use the same target trajectory, random draws, detector settings, and observation schedule. Within each condition, a history predictor will be compared with motion-feature, propagation-feature, and coupled predictors. Complete wave realizations, response regimes, and acquired files will be held out before fitting so adjacent observations cannot appear in both development and test data.

The primary forecast will estimate the probability that the target produces a usable observation on its next scheduled scan, using only history and state estimates available at the forecast time. A secondary forecast will estimate the probability of three consecutive missed visits. The Brier score will be the primary measure of probabilistic accuracy, with reliability diagrams used to assess calibration. Detection rate, false-alarm rate, missed-run length, outage duration, conditional position-error distributions, and processing time will describe operational performance.

The common forecast label will use one declared position-correction method and tolerance. A paired comparison will separately evaluate midpoint pose, per-azimuth horizontal-motion correction, and per-azimuth attitude correction. This separation ensures that differences in forecast scores reflect differences in predicted usable observations, while the correction comparison measures how mapping choices affect position quality among acquired returns. Misses will remain failures in the position analysis rather than being omitted as if their position error were zero. Appendix D explains the comparison logic, and Appendix D.1 justifies the numerical design.

A Shared wave forcing and mechanical response

Linear Airy theory supplies the shared wave field through small-amplitude superposition and finite-depth dispersion. [6, Sec. 6.2, pp. 3–6] With surface elevation η , component amplitude a_j , propagation vector $\mathbf{k}_j$, angular frequency ω_j , phase ϕ_j , water depth h , and gravity g , the field is

$$\eta(\mathbf{x}, t) = \sum_j a_j \cos(\mathbf{k}_j \cdot \mathbf{x} - \omega_j t + \phi_j), \quad \omega_j^2 = g\|\mathbf{k}_j\| \tanh(\|\mathbf{k}_j\|h). \quad (1)$$

For $\|\mathbf{k}_j\|h \gg 1$, deep water gives $\omega_j^2 = g\|\mathbf{k}_j\|$. A translating vessel follows $\mathbf{x}(t) = \mathbf{x}_0 + \mathbf{U}t$. [7, pp. 1–2] Substitution into Equation 1 then gives the signed encounter frequency $\omega_{e,j} = \omega_j - \mathbf{k}_j \cdot \mathbf{U}$. Head seas increase this frequency because $\mathbf{k}_j$ points along propagation. Speed changes forcing frequency, while fixed oscillator parameters retain the same natural frequency.

The floating-structure equations supply inertia, damping, and restoring terms for a decoupled, small-motion approximation. [8, pp. 165–167] For roll or pitch mode q , I_q includes added inertia, c_q is damping, k_q is restoring stiffness, and M_q is excitation moment. Oscillator normalization gives [9]

$$I_q \ddot{\theta}_q + c_q \dot{\theta}_q + k_q \theta_q = M_q(t), \quad \omega_{n,q} = \sqrt{k_q/I_q}, \quad \zeta_q = \frac{c_q}{2\sqrt{k_q I_q}}. \quad (2)$$

Heave similarly uses $m\ddot{z} + c\dot{z} + kz = F_w(t)$, with added mass included in m . Local elevation/slope excitation is a project assumption; dispersion does not determine its coefficients. Appendix D.1 supplies numerical choices and their sources.

For harmonic moment amplitude M_0 and $\Omega = |\omega_e|$, Equation 2 gives the complex amplitude $\Theta_q = M_0/(k_q - I_q\Omega^2 + ic_q\Omega)$ by the floating-body harmonic-response construction. [8, pp. 166–167] Normalization by static deflection M_0/k_q gives $H(r) = 1/(1 - r^2 + 2i\zeta r)$, with $r = \Omega/\omega_n$. Dimensionless magnification and forcing-relative phase lag follow as

$$|H(r)| = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}, \quad \delta(r) = \text{atan2}(2\zeta r, 1 - r^2). \quad (3)$$

Thus angular amplitude is $(M_0/k_q)|H(r)|$; wave-to-motion response additionally requires the excitation coefficient.

Figure 1 connects the shared wave forcing to vessel dynamics and target–surface kinematics. Acquisition geometry and propagation determine the target echo, which the fixed detector evaluates with clutter and noise. Observation usability combines detection with corrected position quality, so acquisition geometry also feeds the output directly.

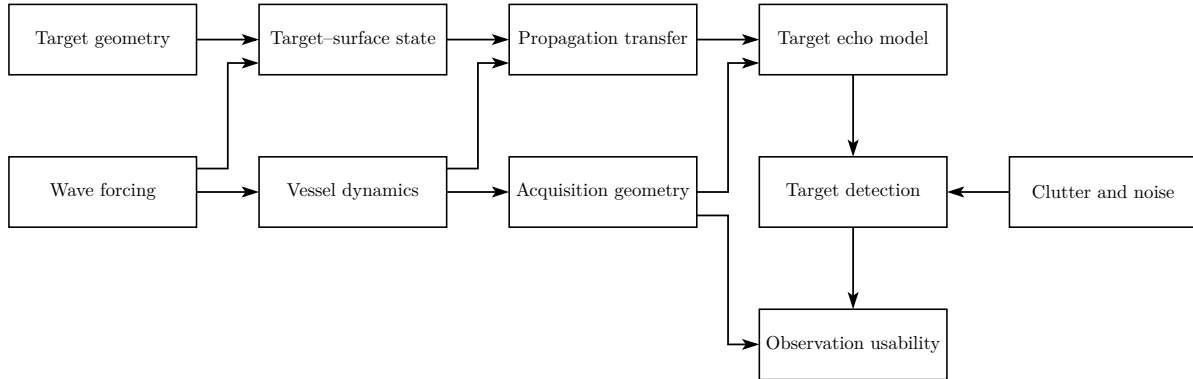


Figure 1: Shared-wave model for usable target observations.

B From ray-time geometry to a usable observation

Mechanical response determines the radar origin and rotation at acquisition time t_i . With vessel reference position $\mathbf{p}_i$, body rotation $\mathbf{R}_{b,i}$, lever arm ℓ , and radar extrinsic rotation $\mathbf{R}_{br}$, use $\mathbf{s}_i = \mathbf{p}_i + \mathbf{R}_{b,i}\ell$ and $\mathbf{R}_i = \mathbf{R}_{b,i}\mathbf{R}_{br}$. Slant range ρ_i and azimuth α_i define a family indexed by unmeasured elevation ϵ ,

$$\mathbf{q}_i(\epsilon) = \mathbf{s}_i + \rho_i \mathbf{R}_i \begin{pmatrix} \cos \epsilon \cos \alpha_i \\ \cos \epsilon \sin \alpha_i \\ \sin \epsilon \end{pmatrix}. \quad (4)$$

A known synthetic target-height plane closes this inverse. Public two-dimensional returns retain an elevation interval or target-height assumption. A general angular displacement bound cannot substitute for the horizontal error of this geometry. Figure 2 shows how radar and target motion change beam offset, direct-path clearance, and mapping pose. These changes produce distinct obstruction, beam-loss, and position-error mechanisms.

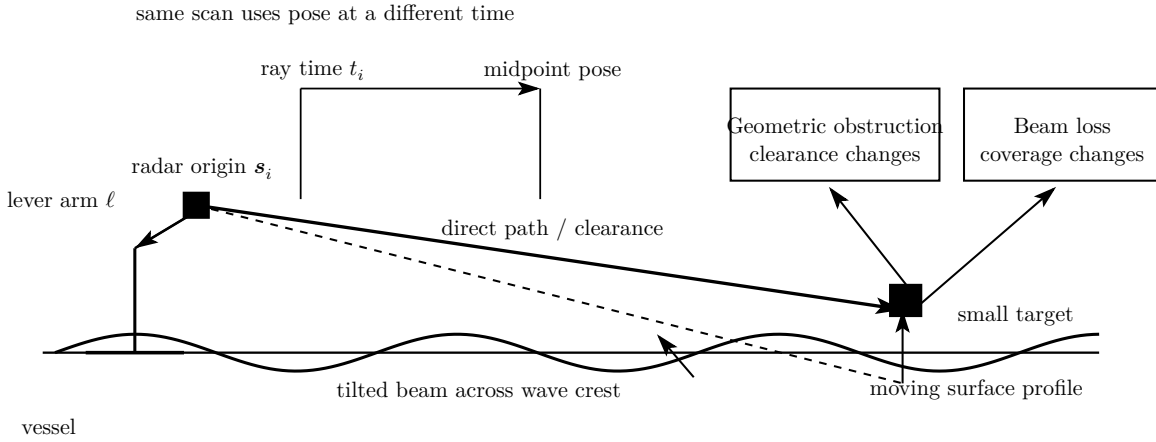


Figure 2: Acquisition geometry and wave-path obstruction.

Let B_i be one-way antenna power gain at the target direction and G_i the one-way propagation power factor. The illustrative Gaussian beam uses full half-power width β and $B_i = \exp[-4 \ln(2)(\epsilon_i/\beta)^2]$. For an obstructing edge of height d_c above the direct ray, wavelength λ , and path distances d_1, d_2 , the limiting diffraction parameter is

$$\nu = d_c \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}}, \quad J(\nu) = 6.9 + 20 \log_{10} \left(\sqrt{(\nu - 0.1)^2 + 1} + \nu - 0.1 \right). \quad (5)$$

The illustration sets $J = 0$ for $\nu \leq -0.78$ and $G = 10^{-J/10}$. The maximum normalized intrusion along the path represents one dominant obstacle. This ITU-R approximation supplies a check and sensitivity arm; a reflective smooth sea crest requires richer propagation treatment. [10]

For baseline target power $P_{0,i}$ and reciprocal complex propagation q_i with $|q_i|^2 = G_i$, a monostatic return is $Y_i = \sqrt{P_{0,i}} B_i q_i^2 e^{i\phi_{T,i}} + C_i + N_i$. Thus target power scales as $P_{0,i} B_i^2 G_i^2$. Target phase, clutter C_i , noise N_i , range loss, polarization, and integration are explicit terms. Incident illumination alone is not a detection. [3]

For detector decision D_i , correction method $\mathcal{C}$, horizontal error $E_i(\mathcal{C})$, and tolerance $E_{\max}$, define usable observation $U_i(\mathcal{C}) = D_i \mathbf{1}\{E_i(\mathcal{C}) \leq E_{\max}\}$. The probability forecast uses one frozen $\mathcal{C}$ so all predictors share labels. Separate correction experiments compare U_i across mapping methods. Missing returns remain misses; they do not become zero-error positions.

C Published-model calculations

The benchmark has two independent literature baselines. Wijaya and van Groesen construct surface visibility from geometric obstruction. Their published harmonic case uses period $T = 9$ s, depth $h = 50$ m, and radar-height ratios $H_r/A = 5$ and 10. [11] The published chapter gives the obstruction indicator and its image average as Equations 3.4 and 3.7. [12] With a surface point $\eta(r, t)$ and radar height H_r , the equivalent skyline test is

$$\chi(r, t) = \mathbf{1} \left\{ \frac{\eta(r, t) - H_r}{r} \geq \sup_{0 < x < r} \frac{\eta(x, t) - H_r}{x} \right\},$$

$$V(r) = \frac{1}{M} \sum_{j=1}^M \chi(r, t_j). \quad (6)$$

For $\eta = A \cos(kr - \omega t)$, exact finite-depth dispersion gives $\lambda = 124.829$ m. The reproduction uses the dissertation’s Figure 3.4 case and averages $M = 1400$ uniform phases on a grid of $\lambda/400$. It reproduces the paper’s numerical visibility construction rather than digitizing its graph. In Figure 3, visibility at $r/\lambda = 10$ is 0.153 for $H_r/A = 5$ and 0.235 for $H_r/A = 10$. Doubling phase and radial resolution changes the curves by at most 0.0011. This surface-visibility baseline characterizes propagation rather than target detection probability.

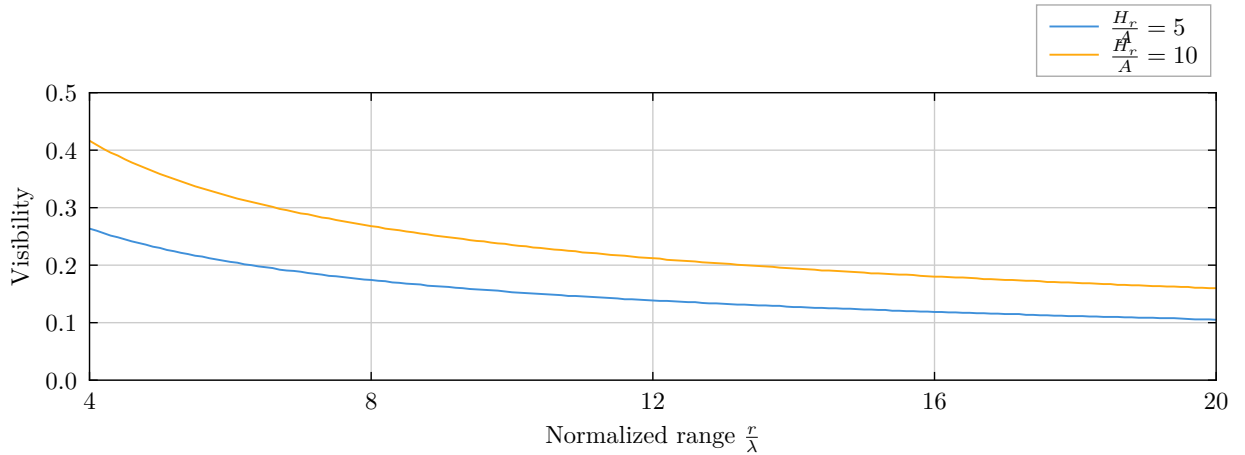


Figure 3: Calculated surface visibility for two normalized radar heights.

Lund et al. map each pulse with interpolated horizontal vessel position and heading. Their radar rotates in 1.25 s and uses a 7.5 m Cartesian grid. [4] In the constant-heading limit, acquisition-time mapping is $\mathbf{x}_i = \mathbf{p}(t_i) + \mathbf{R}(\psi_i)\mathbf{r}_i$. Holding pose at the midpoint instead introduces $E_h = U|t_i - t_{\text{mid}}|$. Using the paper’s 11 kn speed example gives a maximum midpoint error of 3.537 m in Figure 4. Ideal ray-time mapping removes the prescribed translation error, while the 0.5 m reference marks the proposed position tolerance. These calculated errors are model outputs, rather than Lund et al.’s measured residuals.

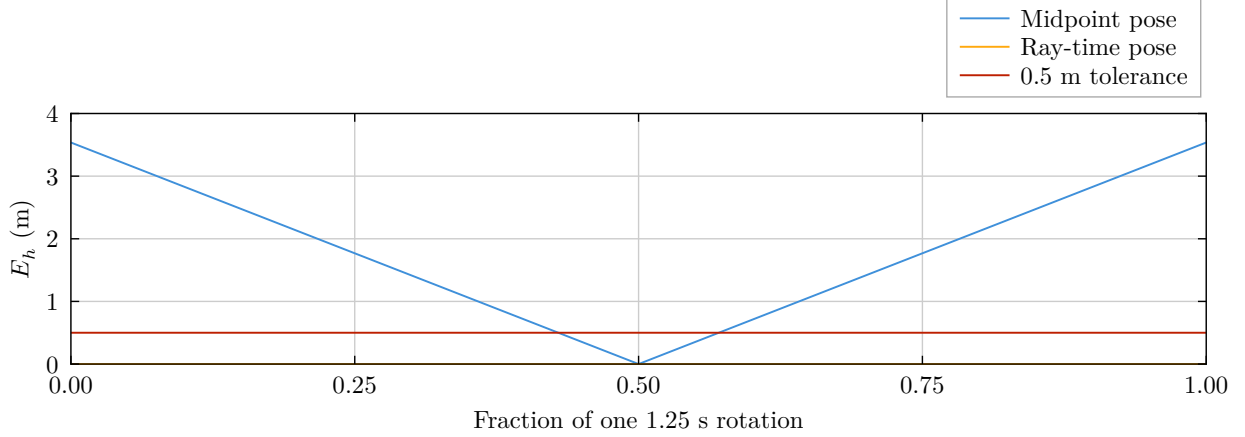


Figure 4: Calculated localization error for midpoint and ray-time mapping.

Together, these baselines separate surface shadowing from acquisition distortion before coupling them. The semester study adds vessel oscillation, target exposure, beam gain, and a received-echo model, then tests whether the combined state improves a common forecast event.

The two model reproductions will serve as component checks before the coupled benchmark. The visibility curve tests the obstruction geometry and phase average, while the constant-translation result tests acquisition-time mapping without an oscillator or detector. Consequently, agreement with either baseline verifies a specific implementation component; it does not measure forecast skill. The IPIX record supplies a separate example of measured return variability. Its use here documents preparation evidence and does not select it as the project’s experimental evaluation dataset.

C.1 Measured small-target radar fluctuations

The calculations above establish mechanisms, while an actual radar acquisition establishes the processing problem. IPIX record 17 contains a one-meter mesh-wrapped floating sphere, with primary range bin 9 and secondary bins 8-11. [13] The downloaded record has 131,072 sweeps, 14 range bins, 9.39 GHz carrier frequency, a 30 m antenna height, and bin 9 beginning at 2694 m. Horizontal-transmit/horizontal-receive (HH) data are processed below.

The first 32,768 sweeps train one common receiver correction using clutter bins 1-7 and 12-14. Unsigned-byte interpretation follows the official loader. Separate in-phase and quadrature means and scales, followed by phase-imbalance correction, are learned from that clutter and applied identically to every bin. [14] This retains target-to-clutter contrast. For corrected samples I_n, Q_n , nonoverlapping 64-sweep powers are $P_j = \frac{1}{64} \sum_{n=64j}^{64j+63} (I_n^2 + Q_n^2)$. The development-clutter mean sets the relative dB scale; its 99th percentile sets the fixed gate at 11.346 dB.

Figure 5 shows the held-out target and clutter powers under this frozen correction and gate. The file and tutorial use conflicting pulse-frequency conventions, so the horizontal axis retains sweep index. Power is expressed relative to the development-clutter mean, rather than as calibrated target radar cross section.

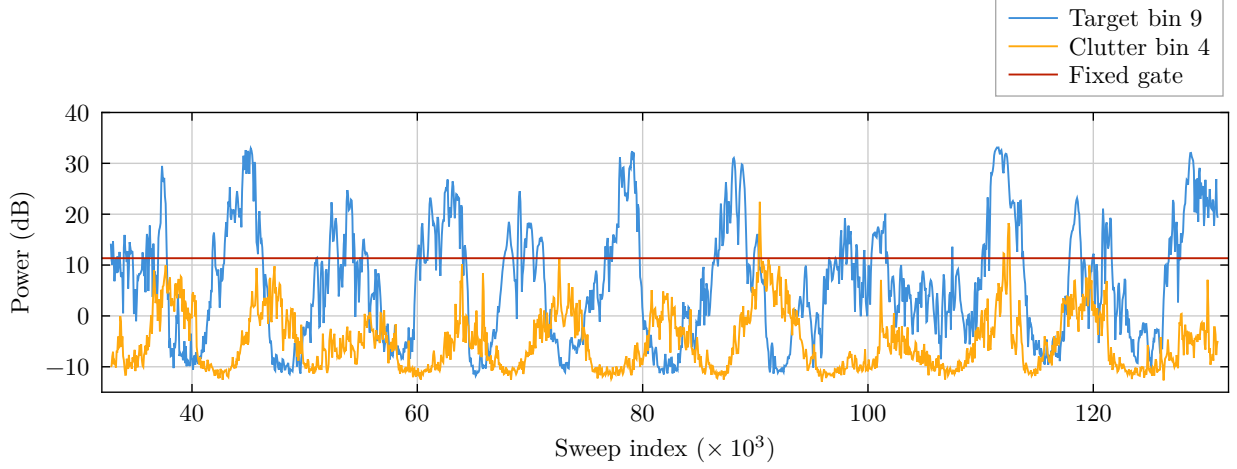


Figure 5: Held-out target-bin and clutter-bin power in IPIX record 17.

Figure 6 compares exceedance fractions across candidate power thresholds, with the vertical reference marking the frozen development gate. Clutter is pooled from ten non-target bins. The target-bin crossings provide a power-based observability proxy; the record does not provide per-sweep target-visibility labels.

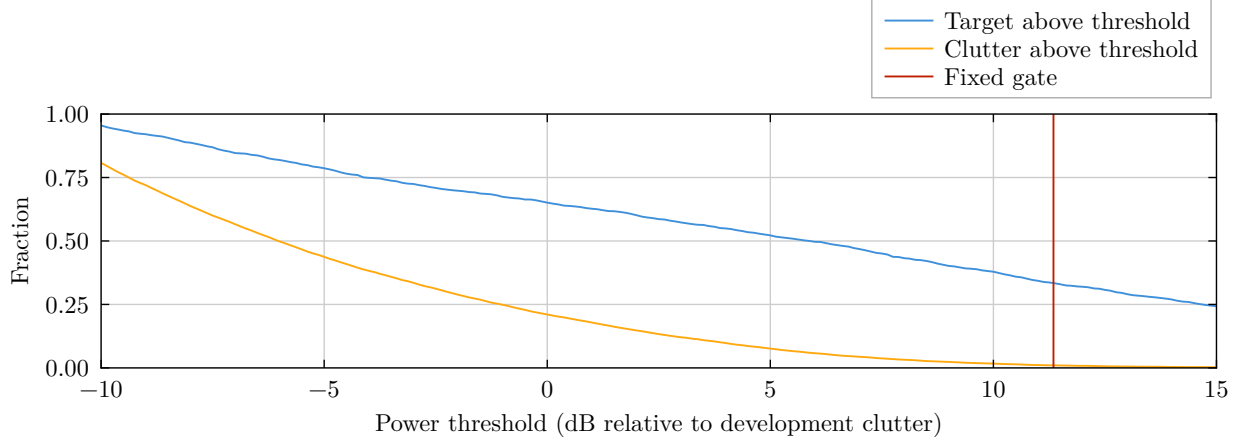


Figure 6: Held-out power exceedance fractions in IPIX record 17.

Table 1: Measured power-gate quantities in the held-out IPIX segment.

Measured quantity	Value	Measured quantity	Value
Held-out primary windows	1536	Held-out clutter windows	15,360
Clutter above fixed gate	0.990%	Primary bin below fixed gate	66.602%
Longest primary run below gate	6656 sweeps	Nonfinite / missing samples	0

Because target-bin power crosses the same calibrated gate repeatedly, average return power alone cannot describe persistence. These measurements motivate an outage-aware forecast. They do not identify a wave cause. Local wave profiles and target pose remain necessary for that attribution, and a bin-wise power gate remains distinct from the proposed spatial detector.

D Benchmark rationale and comparison design

The central question is whether a shared estimate of waves and vessel response predicts the next usable target observation better than detection history or either mechanism alone. A comparison of unrelated scenes cannot answer that question because a change in sea state also changes target motion, clutter, and acquisition geometry. Thus, the benchmark separates changes to the physical generator from changes to the information supplied to the predictor. The former identifies which mechanisms cause loss; the latter tests whether estimating those mechanisms improves a forecast of the same observations.

Four physical conditions will be generated from each underlying realization. The target trajectory, wave phases, sensor configuration, scan clock, and background random draws remain common. The reference condition freezes the radar at its nominal pose and sets path transmission to unity. Motion only enables vessel motion and its effects on pointing and mapping. Propagation only enables wave-path attenuation while keeping the radar pose nominal. The coupled condition enables both. Target motion remains common to all four conditions, so changing a physical arm does not quietly substitute a different target experiment. Table 2 states the interventions explicitly.

Table 2: Paired physical conditions and the question each answers.

Condition	Enabled changes	Interpretation
Reference	Nominal radar pose and unit path transmission; the common target still moves.	Provides the shared target, receiver, and background reference.
Motion only	Time-resolved vessel pose, lever arm, antenna pointing, and mapping; unit path transmission.	Measures loss and distortion introduced by platform motion.
Propagation only	Nominal radar pose with evolving wave-path transmission.	Measures path loss independently of vessel pointing.
Coupled	Time-resolved pose and transmission from the same spatial wave realization.	Tests the joint outcome when the two mechanisms share forcing.

Pairing also makes the numerical comparison more efficient. If two score estimates are S_a and S_b , then $\text{Var}(S_a - S_b) = \text{Var}(S_a) + \text{Var}(S_b) - 2\text{Cov}(S_a, S_b)$. Common random numbers reduce comparison variance when they induce positive covariance; their benefit is therefore checked rather than assumed. [15] The primary ablation will hold the background process common to isolate target-path and antenna effects. However, because real sea clutter can also change with waves and look direction, a separate sensitivity comparison will vary its conditional distribution. Otherwise, an apparent propagation benefit could simply reflect an unrealistically fixed clutter field.

Within each physical condition, four predictors will use the information sets in Table 3. The history predictor establishes how much persistence alone explains. Motion features add estimated vessel state and acquisition geometry. Propagation features add estimated wave and target state. The coupled predictor adds both groups and a small set of declared phase, pointing, and transmission interactions. Each predictor will use logistic regression with the same regularization-selection and probability-calibration procedure. Greater input dimension is allowed, but a different model family or a larger tuning search is not introduced only for the coupled case. Ergo, the comparison tests added physical information under a common learning procedure.

Table 3: Predictor inputs evaluated on identical observations within each physical condition.

Predictor	Available information	Reason for comparison
History	Recent detections, local clutter power, range, and observable target-context features.	Tests persistence and radar context without an estimated physical state.
Motion	History plus estimated attitude, angular rate, radar-origin motion, beam offset, and acquisition timing.	Tests whether acquisition-state information adds forecast value.
Propagation	History plus estimated wave phase, target heave, path clearance, and transmission.	Tests whether wave-path information adds forecast value.
Coupled	Both state groups and declared interactions among phase, beam offset, and transmission.	Tests added value over both single-mechanism predictors.

The simulator will retain complete physical truth, but the predictors will receive only estimates available at the forecast cutoff. A perfect-state predictor can be reported as an oracle diagnostic, separately from the operational comparison. All state errors will be applied before generating predictor inputs; future target position, future waves, and test outcomes will not be exposed during fitting. Future states may be propagated from the cutoff estimates using frozen dynamics, while subsequent simulator truth remains reserved for evaluation. Thus, knowing the synthetic truth does not automatically give the forecaster the answer.

Let D_i denote a correctly associated target detection, and let $E_i(\mathcal{C}_0)$ be its horizontal position error under one frozen correction method $\mathcal{C}_0$. With tolerance $E_{\max}$, the next-scan event follows the usable-observation definition in Appendix B. The primary probability p_i and the separate three-miss event V_i are

$$p_i = \Pr(U_{i+1}(\mathcal{C}_0) = 1 \mid \mathcal{F}_i),$$

$$V_i = \prod_{j=1}^3 (1 - D_{i+j}), \quad q_i = \Pr(V_i = 1 \mid \mathcal{F}_i). \quad (7)$$

Here $\mathcal{F}_i$ contains only history and estimated state available at the cutoff. The next visit is defined by its azimuth acquisition time, rather than by the scan midpoint. The three-miss probability will be fitted as a joint sequence event; multiplying three marginal miss probabilities would impose independence precisely where waves create temporal dependence. A three-visit outage horizon is a proposed tracking stress case, while the next-scan horizon is the main research question.

The Brier score measures the quality of these probabilities before any warning threshold is chosen. Using N eligible forecasts, its binary form is

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N [p_i - U_{i+1}(\mathcal{C}_0)]^2. \quad (8)$$

The same calculation scores q_i against V_i . Brier introduced probability-based forecast verification, and the proper-scoring framework explains why the expected score is minimized by the true event probability. [16, 17] However, a favorable average can hide a predictor that mainly repeats the event prevalence. Therefore, reliability diagrams will show observed frequencies within forecast-probability groups, and a constant probability estimated on development data will accompany the history baseline. Results will be reported within each physical condition and held-out regime rather than averaged across conditions that contain different amounts of usable data.

Table 4: Evaluation outputs and the inference each supports.

Output	Measurement	Why it is required
Probability quality	Paired Brier differences and reliability for p_i and q_i .	Distinguishes forecast improvement from an arbitrary alarm threshold.
Detector behavior	True-target detection fraction and clutter-only false-alarm fraction at the frozen operating point.	Checks that a forecast gain was not obtained by changing detector sensitivity.
Persistence	Consecutive missed visits, outage duration, and event counts.	Distinguishes isolated weak returns from sustained sensing interruptions.
Position quality	Median and 95th-percentile error among acquired target returns, paired with usable fraction and miss fraction.	Prevents selective target loss from appearing as improved localization.
Operational cost	Processing time per scan and per forecast.	Tests whether the predictor can finish before the next scheduled observation.

Correction methods will be compared in a separate experiment using the same detected target associations and paired scenes. Midpoint pose, per-azimuth horizontal motion, and per-azimuth attitude treatment will each map the acquired returns, following the acquisition-time registration problem studied by Lund et al. and Burnett et al. [4, 5] The forecast labels remain based on $\mathcal{C}_0$. In contrast, the correction experiment will report how each mapping changes conditional error and usable fraction. Position error is undefined for a missed target, so misses will count against joint usability and be reported alongside conditional errors; no zero error will be assigned to an absent return.

Complete wave realizations and acquired files will be assigned to development, calibration, or test partitions before any fitting. Within the synthetic test, selected frequency and response regimes will also remain outside development. Roberts et al. show why random splitting of structurally dependent observations can underestimate prediction error. [18] Here, adjacent scans share waves and target dynamics, so a random scan split would reward recognition of a previously seen trajectory. Probability calibration, regularization, feature scaling, and detector choices will therefore be frozen before opening the test partition. An experimental dataset has not yet been selected; this grouping rule will be applied to the independent acquisition unit supported by that dataset.

The primary comparison will use realization-level paired differences between the coupled predictor and the history, motion, and propagation predictors. Bootstrap resampling will draw complete independent realizations, keeping all scans and paired methods together. This applies Efron’s resampling principle to the independent experimental unit rather than treating correlated scans as independent observations. [19] Intervals will be computed within held-out regimes. Evidence for the proposed contribution requires consistent improvement over both single-mechanism predictors, with uncertainty excluding zero for the declared paired comparisons. A gain over history alone would show that physical features help, but would not establish that their coupling is necessary. Thus, the main inference comes from paired forecasts on common labels, rather than from a score contrast whose value depends on the chosen scale.

D.1 Parameter design and numerical justification

The benchmark above determines what must change, while parameter selection determines whether those changes are physically interpretable. Published models supply dispersion, response structure, and acquisition reference cases. The numerical ranges below are proposed study settings chosen to span those mechanisms. They will be filtered by the governing equations and convergence checks, rather than treated as independent boxes of plausible numbers. Literature-derived reference values and calculated constraints are identified separately in the tables.

D.1.1 Wave field and admissible amplitude

Linear wave theory expands about a small free-surface slope and relates frequency, depth, and wavelength through the dispersion relation. [6] Thus, amplitude cannot be selected independently of frequency. With

$g = 9.81 \text{ m s}^{-2}$, the deep-water screening calculation is

$$k = \frac{(2\pi f)^2}{g}, \quad \lambda = \frac{2\pi}{k}, \quad ka \leq 0.20 \implies a \leq \frac{0.20}{k}. \quad (9)$$

The cutoff 0.20 is the proposed linear-model eligibility screen, not a universal breaking threshold. Table 5 retains the intended amplitude and frequency ranges, and Table 6 shows why some combinations must be removed. At finite depth, the full relation in Appendix A will be solved before screening. For a multicomponent surface, the realized combined slope and exposure will also be checked; satisfying the component screen alone does not ensure a linear combined field.

Table 5: Wave inputs and their source or design rationale.

Quantity	Initial selection	Basis and purpose
Amplitude a	0.10, 0.30, and 0.50 m before filtering.	Proposed low-amplitude test levels; joint ka filtering preserves the declared linear regime.
Frequency f	0.15, 0.25, 0.35, and 0.50 Hz.	Proposed range of periods 6.67–2.00 s spanning the initial response study.
Depth h	50 m for the initial finite-depth reference.	The reproduced Wijaya harmonic case uses this depth; other depths require the exact dispersion and model checks. [12]
Heading and speed	Head, following, and oblique seas; $U = 0\text{--}1 \text{ m s}^{-1}$.	Use $\omega_e = \omega - \mathbf{k} \cdot \mathbf{U}$ to change forcing frequency consistently. [20]
Phase and direction	Uniform initial phase on $[0, 2\pi)$, shared across vessel and target.	The synthetic-wave construction retains spatial phase, so body motions are not sampled independently. [12]

Table 6: Calculated deep-water wavelength and maximum amplitude under the proposed $ka \leq 0.20$ screen.

f (Hz)	λ (m)	$a_{\max}$ (m)	Admissible proposed amplitudes
0.15	69.392	2.209	0.10, 0.30, 0.50 m.
0.25	24.981	0.795	0.10, 0.30, 0.50 m.
0.35	12.745	0.406	0.10, 0.30 m.
0.50	6.245	0.199	0.10 m.

For example, $a = 0.30 \text{ m}$ at $f = 0.35 \text{ Hz}$ gives $ka = 0.1479$ and passes the screen. However, $a = 0.50 \text{ m}$ at the same frequency gives $ka = 0.2465$ and fails. At 0.50 Hz, even $a = 0.30 \text{ m}$ gives $ka = 0.3018$. These exclusions are consequential. Without them, a high-frequency failure could be attributed to radar behavior even though the assumed wave model had already left its declared regime. The published $T = 9 \text{ s}$ case in Appendix C remains a separate reproduction check rather than another observation from the proposed operating distribution.

D.1.2 Mechanical response and resonance coverage

The response study will emphasize the frequency ratio $r_e = |\omega_e|/\omega_n$, rather than assign every vessel the same natural frequency. Damped oscillator theory distinguishes inertia, damping, and restoring stiffness, while the response equation in Appendix A gives the harmonic gain and phase. [9] The initial absolute range $f_n = 0.25\text{--}0.75 \text{ Hz}$ is a study bracket. Configurations will be selected to include r_e near 0.5, 1, and 2, subject to compatible wave and speed settings, so the benchmark samples both sides of resonance as well as resonance itself. Measured vessel parameters can later replace these coordinates without changing the comparison protocol.

Table 7: Mechanical response variables and the numerical behavior they test.

Quantity	Initial selection	Justification
Natural frequency	$f_n = 0.25\text{--}0.75$ Hz, with below-, near-, and above-resonance cases.	The ratio r_e determines response gain; absolute frequencies are proposed until a vessel is identified.
Damping ratio	$\zeta = 0.10, 0.15, 0.25$, and 0.40 .	At $r_e = 1$, gain $1/(2\zeta)$ equals 5.00, 3.33, 2.00, and 1.25, respectively. These values expose amplitude and phase sensitivity.
Static pitch forcing	$0^\circ\text{--}3^\circ$ equivalent static deflection.	Include a zero-forcing check and finite excitation. At $\zeta = 0.10$, 3° excitation gives 15° resonant amplitude, so eligibility must enforce the small-angle assumption.
Roll coupling and phase	Roll forcing fraction $0\text{--}0.8$ of pitch; relative forcing phase -180° to 180° .	Include zero roll and varied alignment of pointing loss with wave-path loss.
Target response	Separate target-heave gain and phase from vessel response.	The same surface forcing need not give the same body motion. Target exposure is checked throughout each retained realization.

The initial small-angle screen will retain peak roll and pitch no greater than 10° . This is a numerical model choice justified by the approximation error: $\sin \theta / \theta$ differs from unity by about 0.51% at 10° . Thus, the lowest-damping, largest-excitation corner will be reduced or excluded before interpreting its radar effect. Tightening the limit to 5° will test sensitivity to that choice. At $r_e = 1$, the normalized response must give gain $1/(2\zeta)$ and a 90° displacement lag. The amplitude maximum can occur below $r_e = 1$ when damping is nonzero, so this check is not confused with the frequency of the resonant peak.

D.1.3 Acquisition, geometry, and uncertainty

The scan period controls how much the forcing phase changes while an image is assembled. The proposed periods $T_s = 0.125, 0.25$, and 0.50 s will be supplemented by Lund et al.’s documented 1.25 s instrument case. [4] At fixed-position forcing frequencies $0.15\text{--}0.50$ Hz, these periods span phase advances of $6.75^\circ\text{--}22.5^\circ$, $13.5^\circ\text{--}45^\circ$, $27^\circ\text{--}90^\circ$, and $67.5^\circ\text{--}225^\circ$, respectively. The actual moving-vessel advance is $|\omega_e|T_s$. Ergo, scan time is a physical parameter tied to wave response, rather than simply a plotting interval.

Table 8: Acquisition and geometry settings chosen to expose scan and beam effects.

Quantity	Initial selection	Derived scale or reference
Scan period	0.125, 0.25, 0.50, and 1.25 s.	The first three are proposed stress settings; 1.25 s is the published shipboard reference. [4]
Range	$R = 20, 50, 100$, and 300 m.	Translation distortion scales with UT_s , while angular coverage and height-plane sensitivity depend on range. These ranges are proposed geometry levels.
Radar and target height	Radar 1–4 m; nominal exposed target 0.10–0.50 m.	Low-height test geometry varies grazing angle and clearance. Each retained target remains exposed above its local surface.
Vertical beamwidth	$\beta = 4^\circ, 8^\circ$, and 12° .	Test target angular offsets below, near, and beyond $\beta/2$. The Gaussian gain in Appendix B gives 6.02 dB two-way loss at the one-way half-power angle.
Pose and lever arm	Nominal sensor offset fixed across paired arms; geometry then repeated with zero offset.	Time, angle, and installation offsets are documented registration concerns. [21]

These values are interpreted through ratios. At $R = 100$ m and $\beta = 4^\circ$, the vertical half-width is $R \tan(\beta/2) = 3.49$ m, while a 3° pointing offset exceeds the 2° half-power angle. Thus, beam coverage can

deteriorate even when the target remains exposed above the water. For translation alone, midpoint error is bounded by $UT_s/2$. At $U = 1 \text{ m s}^{-1}$, the 0.125–0.50 s cases give 0.0625–0.25 m, whereas the 1.25 s case gives 0.625 m. Thus, the scan choices span both sides of the proposed 0.50 m position tolerance even before attitude and lever-arm effects are added.

Table 9: Uncertainty and operating-tolerance cases with explicit scaling.

Quantity	Initial selection	Reason and reporting rule
Position tolerance	$E_{\max} = 0.50 \text{ m}$; sensitivities 0.10 and 1.00 m.	Proposed operational thresholds. Report results at all three; no mission requirement is presumed.
Timestamp error	0, 5, and 25 ms.	At $T_s = 0.25 \text{ s}$ these are 0%, 2%, and 10% of a rotation. Propagate the offset through pose and azimuth association rather than perturbing coordinates directly.
Extrinsic tilt	0° , 0.25° , and 1° .	At 100 m, 1° changes the line-of-sight vertical scale by 1.75 m. This is not automatically horizontal localization error.
Estimated wave state	Relative amplitude error 0%, 10%, and 20%; phase error 0° , 15° , and 30° .	Proposed robustness tests. Phase is varied separately because amplitude error does not represent a mistimed crest.
Target-height estimate	Exact height, then offsets of ± 0.05 and $\pm 0.10 \text{ m}$.	Tests dependence of the elevation closure on prior target geometry. Use identical height assumptions across paired correction methods.

Timing errors must preserve the actual measurement convention. If the radar reports encoder azimuth with an uncertain clock, the error primarily changes the pose assigned to that ray. If azimuth is inferred from scan time, the same clock error can also rotate the assigned bearing by $2\pi\Delta t/T_s$. McCann and Bell treat range, angle, and timing offsets as distinct installation parameters. [21] The simulator will therefore declare which quantity is measured and which is inferred. In contrast, simply adding a range-independent position offset would conceal the mechanism the experiment is meant to explain.

D.1.4 Detector operating point and trial budget

A detector must be common across methods, because changing its threshold would change the target event being forecast. The initial choice is cell-averaging constant false-alarm-rate (CA-CFAR) detection with a development-calibrated scaling factor. Rohling describes local reference-cell thresholding and compares its behavior in uniform clutter, clutter edges, and multiple-target situations. [22] Accordingly, homogeneous exponential-power clutter will provide the analytical check, while compound-Gaussian backgrounds and clutter edges will test robustness. The guard region will be set using development-only target-spread checks before any forecast evaluation. The unattenuated target signal-to-noise ratio (SNR) will be varied while also recording its power margin above the clutter-dependent threshold.

For N_r independent exponential reference powers and a mean-based threshold, the analytical relation is

$$P_{\text{FA}} = (1 + \alpha/N_r)^{-N_r}, \quad \alpha = N_r \left(P_{\text{FA}}^{-1/N_r} - 1 \right). \quad (10)$$

At $N_r = 24$ and $P_{\text{FA}} = 0.01$, this gives $\alpha = 5.077$. This relation checks the implementation under its stated distribution, while the empirical threshold calibration controls the proposed nonhomogeneous cases. The detector uses the same cells, association rule, and scaling procedure for all predictor groups.

Table 10: Detector settings and the evidence used to choose or check them.

Quantity	Initial selection	Justification
Reference window	$N_r = 24$, with 16 as a sensitivity case.	Rohling examines these reference-window sizes. They supply documented starting cases rather than a claim of optimality. [22]
Guard cells	Initially one cell on each side of the cell under test; enlarge if development target spread requires it.	Exclude target leakage from the background estimate, consistent with the two-guard-cell example in Rohling. Freeze the rule before testing. [22]
False-alarm probability	$P_{FA} = 0.01$ per declared decision.	Proposed starting operating point, chosen to permit explicit finite-sample calibration. Report empirical false alarms separately from target misses.
Target-power levels	Unattenuated target SNR 10–25 dB; also record margin above the local threshold.	Proposed power bracket. A 6 dB two-way pointing loss can move a near-threshold target across the decision boundary.
Clutter-only calibration	At least 10,000 independent decisions for the declared initial operating point.	About 100 expected false alarms at $p = 0.01$; precision is quantified below, and test clutter is separate.

The calibration count follows from Bernoulli precision. With $N = 10,000$ independent decisions and $p = 0.01$, the standard error is

$$\text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{N}} = 0.000995, \quad 1.96 \text{SE}(\hat{p}) \simeq 0.00195. \quad (11)$$

Thus, the initial count gives approximately ± 0.002 absolute precision near the design rate; it does not demonstrate an arbitrarily small false-alarm probability. The detector will be calibrated on development clutter and checked again on independent test clutter, with a binomial interval for independent decisions or realization-level uncertainty for clustered decisions. Reusing neighboring correlated cells or evaluating the selected threshold on its own calibration sample would overstate the evidence. Lower operating probabilities require a separately justified larger trial budget.

D.1.5 Configuration coverage and numerical verification

An exhaustive Cartesian sweep grows quickly and wastes computation on repeated combinations far from the failure boundary. A stratified, Latin-hypercube-style design will instead spread the initial configurations over the declared coordinates, following McKay et al.’s sampling approach. [23] Physical constraints will be enforced while generating the design, and the achieved coverage will be recorded after filtering. This matters because rejection can disturb the intended strata. Below-, near-, and above-resonance settings and threshold-crossing beam offsets will be checked explicitly rather than assumed to occur in a small random design.

Table 11: Initial allocation of configurations and independent realizations.

Partition	Initial allocation	Reason and acceptance rule
Development	32 configurations, each with 8 independent phase/background realizations.	Provides 256 paired realizations for fitting and grouped regularization selection; add configurations if mechanism coverage is missing.
Calibration	8 new configurations, each with 8 independent realizations.	Provides 64 paired realizations for forecast-probability calibration without reusing the fitting or test scenes.
Test	16 held-out configurations, each with 8 independent realizations.	Provides 128 paired realizations for comparison across held-out response and wave regimes; uncertainty is based on these groups.
Duration	120 s per realization, with one scheduled target visit per rotation.	At $f \geq 0.15$ Hz this spans at least 18 physical wave cycles. It gives 96–960 visits for the proposed scan periods, not that many independent samples.
Total cost	$56 \times 8 = 448$ underlying realizations and $448 \times 4 = 1792$ physical-condition traces.	A proposed initial computation budget. Expand independent replication if paired intervals remain too wide; extra scan rows alone are insufficient.

These counts allocate effort; they are not a literature-established power calculation. The first pilot will estimate realization-level score variance and detector-event prevalence. Additional independent realizations will then be added when the paired intervals cannot resolve the declared comparison. Numerical resolution is a separate problem. Initial state integration will use at least 50 steps per shortest retained natural or encounter period, with interpolation evaluated at the true ray timestamps. Time steps, path subdivisions, and phase samples will be refined independently, because stable detector fractions alone can hide an unstable geometry calculation.

Table 12: Verification checks and numerical acceptance criteria.

Check	Required behavior	Reason or proposed tolerance
Forcing and response	Zero forcing and zero initial displacement and velocity give zero motion; static gain is one; $r_e = 1$ gives $1/(2\zeta)$ gain and 90° lag.	Analytical checks of the oscillator equations in Appendix A.
Frame and plane inverse	Known pose and target plane recover the synthetic target; zero lever arm removes its displacement.	Validate transformation order before interpreting correction errors.
Beam and echo	At $ \epsilon = \beta/2$, one-way power gain is $1/2$ and reciprocal two-way beam gain is $1/4$.	Validate the distinction between incident illumination and received target power.
Detector	The exponential-clutter reference recovers Equation 10 within its sampling interval.	Check threshold scaling separately from propagation and forecast quality.
Discretization	Halve the state time step and double path and phase resolution in separate checks.	Proposed acceptance changes no more than 0.01 in usable fraction, 0.005 in Brier score, and $0.1E_{\max}$ in the 95th-percentile geometry error.
Information boundary	Future truth is unavailable to fitted predictors; paired methods share labels; missing returns remain unusable.	Reject leaked features or selective target filtering before score comparison.

The proposed convergence tolerances keep numerical changes smaller than the operational and statistical comparisons being reported. If a claimed predictor gain is smaller than the Brier refinement tolerance, refinement must continue before interpreting that gain. Finally, binary obstruction and continuous diffraction transmission will be evaluated as alternate propagation assumptions, with the latter tied to the limiting model

in Appendix B. These controls establish whether the conclusion survives the observation model instead of merely reflecting the preferred simulator.

E Related Works

The preceding benchmark combines established wave, propagation, detection, and mapping models. The relevant literature therefore needs to be read by the quantity each study predicts or measures. Surface visibility, received backscatter, a detector decision, and geographic position are different outcomes; the proposed forecast connects them through a common state and a declared usable-observation event.

E.1 Small-target detection and sea clutter

The sea-clutter literature makes the target decision conditional on the received field. Panagopoulos and Soraghan’s three processing families suppress unwanted echo structure through averaging, time-frequency separation, and morphological filtering, and their real-data examples show that small-target extraction is feasible under clutter variability. [1] However, these operations do not supply a physical explanation for a missed visit. A low detector response may result from clutter masking, target aspect, propagation loss, antenna gain, or a target return arriving between useful beam samples. The proposed model therefore carries the target echo, sea clutter, and receiver noise as separate terms before applying a fixed detection rule. This formulation preserves the distinction between an unobserved target and an observed target whose geographic position is inaccurate. It also makes the forecast label operational. A usable observation requires both a detector decision and a position error below the declared tolerance.

E.2 Wave shadowing, partial illumination, and diffraction

Wave geometry supplies an important propagation baseline, but the evidence does not support treating visibility as a binary law. Plant and Farquharson distinguish geometric shadowing, in which incident rays are completely blocked, from partial shadowing, in which diffraction and weak scattering can leave a reduced return in a geometrically shadowed region. Their coherent X-band measurements test how the fraction of near-noise samples changes with incident power and polarization. The observed behavior does not support geometric shadowing as a dominant explanation for low-grazing-angle deep-water microwave backscatter, while polarization-dependent partial-shadowing effects become more relevant for steep waves near shorelines. [2] The implication is direct. A geometric line-of-sight indicator is useful as a sensitivity baseline, yet it cannot be interpreted as target detection probability or as proof that a return is absent. The received target term must retain propagation attenuation, clutter, noise, polarization, and detector-threshold effects.

Wijaya and van Groesen use a synthetic radar-image construction to relate ocean-surface visibility to geometric obstruction. Their analysis gives a deterministic connection between normalized radar height, range, wave elevation, and the fraction of a surface record that remains visible. [11] The associated dissertation reproduces the obstruction indicator and image-average construction in its third chapter, including the harmonic parameter cases used to examine significant wave height and wave reconstruction. [12] These works provide a useful geometry-first reference because a shared wave field can be evaluated at the radar and target locations with a common phase. However, surface visibility is an intermediate state. It does not include target radar cross section, antenna beam gain, two-way propagation, sea clutter, receiver noise, or the scan-time decision that determines whether a target is detected. The present study consequently uses geometric obstruction to expose a mechanism and then propagates the same state into an explicit received-echo model.

Diffraction sharpens the distinction between incident illumination and a usable target observation. Janssen et al. calculate diffraction effects in radar illumination of sea waves and show that the field near and beyond a wave-shadow boundary can differ from a geometric-optics construction. [3] Their result supports a diffraction sensitivity arm around the geometric baseline. It does not by itself establish a target-return or detection model, because illumination is only one factor in the monostatic echo. Ergo, the relevant forecast state must include the target path, the two-way propagation factor, antenna response, and clutter and noise conditions under which the detector operates. This coupling also avoids attributing every missed observation to wave blockage when partial illumination or weak scattering provides a competing explanation.

E.3 Wave-driven mechanical response

The waves that change propagation also excite the radar vessel. Linear wave-body theory connects incident waves to body motion through hydrodynamic forcing, restoring forces, inertia, and damping. Its response-amplitude operators retain frequency-dependent gain and phase, while encounter frequency accounts for vessel speed and heading. [20] Consequently, speed changes the forcing experienced by the vessel even when its prescribed natural frequencies remain fixed. Wave phase controls both the obstructing surface and radar attitude; independently drawing those states would discard the relationship the project aims to test.

The initial damped modes preserve resonance and phase lag while keeping the semester study tractable. Their swept effective parameters are design coordinates. Once a vessel and acquisition are selected, measured response-amplitude operators can replace them. Thus, the benchmark tests shared forcing before vessel-specific identification, while retaining the same forecast event and paired comparisons.

E.4 Scan-time motion and correction

A rotating radar does not acquire a scan at one instant. Each azimuth sample is associated with its own acquisition time, sensor position, orientation, and beam direction. Lund et al. demonstrate pulse-by-pulse georeferencing for shipboard marine radar by combining time-tagged radar measurements with interpolated vessel position and heading. Their radar completes a rotation in 1.25 s and maps pulses on a 7.5 m Cartesian grid. Heave, pitch, and roll are omitted because their effects are judged small relative to that resolution and expected to average out. Thus, the work supplies a horizontal-motion baseline and a stated scale argument, rather than evidence that vertical motion can always be ignored. [4] McCann and Bell address a related registration problem for ship-borne X-band intensity imagery. They identify finite angular, range, and timing offsets introduced by equipment installation and estimate those offsets by maximizing the sharpness of time-integrated imagery formed from radar images and high-frequency vessel navigation data. [21] Together, these studies establish that installation and acquisition errors can degrade the location of an acquired return. The proposed correction comparison will use their time-aware mapping principle, while separately perturbing the vessel attitude and radar-to-navigation lever arm.

Spinning-radar navigation work makes the scan-distortion mechanism explicit. Burnett, Schoellig, and Barfoot show that motion distortion and Doppler effects can materially affect radar odometry, and they evaluate an estimator that accounts for both effects on spinning-radar scans. [5] Their experiments concern automotive radar navigation, so the platform and scene differ from a wave-driven maritime vessel. The transferable result is the acquisition principle. Motion during a scan changes the geometry represented by the assembled image, and a midpoint-pose approximation needs to be tested against the time structure of the scan. For the present proposal, vessel heave, pitch, roll, translation, and sensor offsets are evaluated at each azimuth time. Correction quality is then assessed separately from the physical processes that prevent a target echo from crossing the detector threshold.

E.5 Research gap and proposed connection

These studies supply complementary pieces of a reliability model. Detection work addresses extraction from clutter; shadowing and diffraction work describes illumination; shipboard and spinning-radar studies address acquisition-time registration. However, their reconstruction, detection, and navigation objectives do not individually define the joint next-scan event. The benchmark tests whether connecting these states improves that forecast, with novelty assessed against the resulting evidence and final literature comparison.

Thus, the contribution is an interpretable forecast using a common wave realization to connect body response, target state, propagation, beam gain, and scan geometry. The predictor receives history and state estimates available before the next scan. The label records whether that scan yields a detected target whose corrected position satisfies the tolerance. This common event tests the predictive value of coupling while retaining separate explanations for propagation loss, clutter masking, beam loss, and position distortion.

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